

Supplemental Materials

July 3, 2026

Table 1: Hyperparameters used in our experiments.

Category	Hyperparameter	Value
LoRA	Rank (r)	16
	Alpha (α)	32
	Dropout	0.05
	Target Modules	q, k, v, o-proj
GRPO	Group Size	8
	KL Coefficient (β)	0.01
	Clip Advantage	3.0
	Epsilon (ϵ)	0.2
Reward Model	Answer Penalty Weight (w_{leak})	0.3
	Structural Penalty Weight (w_{preamble})	0.3
	Factual Weight (w_{grounded})	0.5
	Answer Threshold (τ_{answer})	0.7
	Preamble Threshold (τ_{preamble})	15
Optimization	Policy Learning Rate	1e-5
	Reward Head Learning Rate	1e-5
	Baseline Learning Rate	1e-5
	Weight Decay	0.01
Training	Stage 1 Epochs	2
	Stage 2 Epochs	2
	Batch Size	4
	Training Samples	500
Generation	Temperature	0.7
	Top- p	0.9
	Max New Tokens	768

1 Different Prompts used in the Experiments

Table 2: Sample instruction prompt, user prompt, and response generated by Llama3.1-8B-Instruct.

Prompt: You are a medical reasoning expert. Structure every response as: <code><facts></code> [atomic medical fact] ... <code></facts></code> <code><answer></code> X <code></answer></code> <i>Rules for facts:</i> (1) List 4–8 atomic facts that directly justify your answer. (2) Each fact must be a single, verifiable statement of general medical knowledge. (3) State facts about conditions, drugs, or mechanisms — not about “the patient” or “this case”. (4) Facts must be distinct from each other. <i>Rules for the answer:</i> (1) Output exactly one letter (A, B, C, D, ...) inside <code><answer></code> tags. (2) The answer must be entailed by the facts listed. Scenario: A 67-year-old woman with advanced bladder cancer comes to the physician for a follow-up examination. She is currently undergoing chemotherapy with an agent that forms cross-links between DNA strands. Serum studies show a creatinine of 2.1 mg/dL and BUN of 30 mg/dL. Urine dipstick shows 2+ protein and 1+ glucose. Prior to chemotherapy, laboratory values were within reference range. In addition to hydration, administration of which of the following would most likely have prevented this patient’s current condition? A. Mesna B. Amifostine C. Rasburicase D. Leucovorin Response: <code><facts></code> (1) Cisplatin, a chemotherapy drug that forms DNA cross-links, is known to cause nephrotoxicity. (2) Nephrotoxicity caused by cisplatin results in elevated serum creatinine concentrations. (3) Elevated BUN is often observed in patients with chemotherapy-induced nephrotoxicity. (4) Chemotherapy-induced nephrotoxicity damages renal tubules, leading to proteinuria. (5) Urine dipstick findings of 2+ protein and 1+ glucose indicate tubular kidney damage. (6) Amifostine is a cytoprotective agent that selectively protects normal tissues from cisplatin-induced nephrotoxicity. (7) Mesna prevents hemorrhagic cystitis from cyclophosphamide/ifosfamide but does not prevent cisplatin nephrotoxicity. (8) Leucovorin rescues normal cells from methotrexate toxicity and is unrelated to platinum nephroprotection. <code></facts></code> <code><answer></code>B<code></answer></code>

Table 3: Prompt for Fact Verification

Prompt: """You are a medical fact verification specialist. Evaluate the clinical accuracy of the given atomic fact. VERIFICATION CRITERIA: 1. Medical Accuracy: Is the fact clinically correct? 2. Specificity: Is the claim specific enough to be verifiable? 3. Context Appropriateness: Does the fact make sense in the clinical context? 4. Evidence Base: Is this supported by established medical knowledge? FACT TO VERIFY: {fact_text} MEDICAL CONTEXT: {context} You MUST respond with valid JSON in exactly this format (no extra text): <pre>{{ "confidence_score": 0.85, "is_accurate": true, "reasoning": "detailed explanation of your assessment", "concerns": ["any specific concerns or caveats"] }}</pre> CRITICAL: The confidence_score must be a decimal number between 0.0 and 1.0. ASSESSMENT:"""
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